# Nancy Singh

Software Engineer (TypeScript, Python) – browser testing & AI reliability | Delhi, India | Open to remote and relocation
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## Education

**Sant Longowal Institute of Engineering and Technology (SLIET)** Punjab, India
*B.E. Computer Science and Engineering (lateral entry) – CGPA 7.28/10 (First Division); 8.60 SGPA in semester VII* 2023–2026

**Delhi Skill and Entrepreneurship University (DSEU)** New Delhi, India
*Diploma in Digital Electronics – 66.6%* 2020–2023

## Projects

**Zeitaku** | *TypeScript, TypeScript Compiler API, Playwright, Chrome DevTools Protocol, Vitest* GitHub

- Codemod on the **TypeScript Compiler API** that replaces hard-coded `waitForTimeout` sleeps in Playwright tests with network-aware waits; scanned **7 codebases (76k files)**, found **240** sleeps and left alone the **44** deliberate ones.
- Replaced fixed sleeps with **Chrome DevTools Protocol** request tracking: on throttled Slow 3G, fixed 2 s and 3 s sleeps failed **100%** of runs while Zeitaku failed **0%**; **45% faster** than a safe fixed wait on a normal network.

**Junban** | *TypeScript, Chrome DevTools Protocol, Playwright, delta debugging* GitHub

- Finds the **order of API responses** that breaks a flaky Playwright test by holding each response in the browser (CDP `Fetch`) and releasing them in chosen orders.
- Caught **6/6** planted rare races by the **5th run** vs **2/6 in 3,000** plain reruns; **delta-debugs** each failure to one rule and writes a repro that fails **20/20**.

**Saisei** | *TypeScript, Playwright, network record/replay* GitHub

- Records a web app's API responses once and replays them on every test run; a race-dependent test went from **47/100** passing live to **100/100** on replay.
- Replays concurrent responses in a chosen order to reproduce the failing order **100/100** times, and flags API **drift** by comparing JSON shapes (renamed or retyped fields), with a CLI for CI.

**DocMind** | *Python, FAISS, sentence-transformers, Tesseract OCR, Llama 3.3 (Groq), Streamlit* Live Demo | GitHub

- Deployed a **RAG** app over any PDF (OCR only on pages without a text layer); hybrid **BM25 + embeddings** lifted top-3 retrieval from **88%** to **96%** (scanned pages: 50% → 83%), with a guard that skips the LLM when nothing matches.

**HalluGuard + Kensa** | *Python, PyTorch, Hugging Face, NLI* Live Demo | GitHub | Kensa

- Per-sentence hallucination detector for RAG answers using an **NLI cross-encoder (DeBERTa-v3)**, benchmarked on **RAGTruth**; **Kensa** adds metamorphic tests (reword, distractor, delete evidence) that delta-debug failures to the minimal documents.

## Open Source (11 PRs merged)

**Google Puppeteer** | *TypeScript, Node.js, CDP, WebDriver BiDi* **(5 merged)** #15587 | #15590 | #15595 | #15697 | #15653

- Added `wsOptions.handshakeTimeout` so `connect()` fails fast on a silent host; `page.setGeolocation(null)` removes the override over CDP and BiDi (requested by the maintainer); `request.respond()` skips undefined headers on CDP; BiDi `respond()/abort()` no longer reject for requests that are already gone, and `contentType` now replaces a content-type header. Each with tests.

**Google Chrome DevTools MCP** | *TypeScript, Puppeteer, CDP* **(4 merged)** #3148 | #3144 | #3198 | #3199

- Promise rejections show their real reason; Lighthouse explains unsupported URLs; redirect chains no longer list later hops; console args (Maps, Sets, DOM nodes, cycles) no longer print {}. #3199 was selected for the **XERJ Corpus Hub**, a library AI coding agents search before writing code.

**Hugging Face sentence-transformers** | *Python, PyTorch, learning-to-rank losses* **(1 merged)** #4204

- Fixed `PListMLELambdaWeight` passing the padding mask instead of rank positions to a custom discount fn, which removed the position discount and zeroed the loss of shorter lists; shipped with a failing test.

**Confident AI DeepEval** | *Python, LLM evaluation* **(1 merged)** report #3469 | fix #3470

- JSON datasets saved by DeepEval itself crashed on reload (null `tools_called`); reported and fixed it with a round-trip test.

**Instructor (567 Labs)** | *Python, Pydantic, LLM structured outputs* **(co-author)** #2531 | commit on main

- Proposed a claim-decomposition example for fact-checking pipelines; incorporated into main by the maintainer with co-author credit.

**Bugs found, fixed upstream** | *Microsoft Playwright, Twenty (YC), Documenso* #43459 | #28342 | #3445

- Playwright codegen double-escaped `${`, crashing generated tests; Twenty's Links field dropped slashes in query/hash; Documenso's E2E suite had redundant sleeps (found by Zeitaku). Reported with root causes; fixed by maintainers.

## Skills & Achievements

**Testing & Browser**: Playwright, Chrome DevTools Protocol, Vitest, GitHub Actions, TypeScript Compiler API (AST)
**ML & RAG**: PyTorch, Hugging Face, sentence-transformers, FAISS, LangChain, NLI cross-encoders, TensorFlow Lite
**Languages & Tools**: TypeScript, Python, Kotlin, JavaScript, Git, Node.js, Express, Streamlit, Tesseract OCR, Linux
**Achievements**: RedBus Data Decode Hackathon, Top 500 of 8,000+ teams | NCC "C" Certificate (National Cadet Corps)